Fibonacci Series

Each new number is the **sum of the previous two**: 0, 1, 1, 2, 3, 5, 8, 13, ...

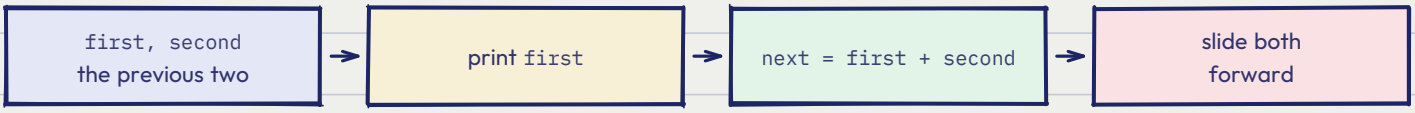
1) Print the first 10 numbers

Example :

```
first = 0
second = 1
for i in range(10):
    print(first, end=" ")
    next_number = first + second
    first = second
    second = next_number
```

Output :

```
0 1 1 2 3 5 8 13 21 34
```



end with a space keeps all the numbers on one line

2) Store them in a list

Example :

```
count = 8
first = 0
second = 1
fibonacci_numbers = []
for i in range(count):
    fibonacci_numbers.append(first)
    next_number = first + second
    first = second
    second = next_number
print(fibonacci_numbers)
```

Output :

```
[0, 1, 1, 2, 3, 5, 8, 13]
```

3) Use a function

Example :

```
def fibonacci(count):
    first = 0
    second = 1
    for i in range(count):
        print(first, end=" ")
        next_number = first + second
        first = second
        second = next_number
fibonacci(7)
```

Output :

```
0 1 1 2 3 5 8
```

Palindrome & Factorial

A **palindrome** reads the same both ways. `[::-1]` makes a reversed copy of a string.

1) Check a word and a number

A word

```
word = "madam"
reversed_word = word[::-1]
if word == reversed_word:
    print(word, "is a palindrome")
else:
    print(word, "is not a palindrome")
```

Output :

```
madam is a palindrome
```

A number

```
number = 121
number_text = str(number)
if number_text == number_text[::-1]:
    print(number, "is a palindrome")
else:
    print(number, "is not a palindrome")
```

Output :

```
121 is a palindrome
```

slicing works on text, so convert a number with str() before reversing it

2) Accept a word from the user

Example :

```
word = input("Enter a word: ").lower()
if word == word[::-1]:
    print("It is a palindrome")
else:
    print("It is not a palindrome")
```

Output :

```
Enter a word: Level
It is a palindrome
```

lower() makes the check ignore uppercase and lowercase differences

3) Factorial (optional)

Example :

```
number = int(input("Enter a positive number: "))
factorial = 1
if number < 0:
    print("Factorial is not defined for negative numbers")
elif number == 0:
    print("Factorial of zero is:", factorial)
else:
    for i in range(1, number + 1):
        factorial = factorial * i
    print("Factorial:", factorial)
```

Output :

```
Enter a positive number: 5
Factorial: 120
```

Star Patterns: Part 1

The **outer loop** controls the rows; the **inner loops** print spaces and stars.

1) print(): sep and end

Example :

```
print("A", "B", sep="-", end=" ")
print("C")
```

Output :

```
A-B C
```

sep goes between the values; end goes after the print and defaults to a new line

2) Increasing and decreasing triangles

Pattern 1 : increasing

```
rows = 5
for i in range(1, rows + 1):
    for j in range(i):
        print("*", end=" ")
    print()
```

Output :

```
*
* *
* * *
* * * *
* * * * *
```

Pattern 2 : decreasing

```
rows = 5
for i in range(rows, 0, -1):
    for j in range(i):
        print("*", end=" ")
    print()
```

Output :

```
* * * * *
* * * *
* * *
* *
*
```

increasing triangle uses range(i); the decreasing one counts the rows down

3) Centered pyramids

Pattern 3 : pyramid

```
rows = 5
for i in range(1, rows + 1):
    for j in range(rows - i):
        print(" ", end=" ")
    for k in range(2 * i - 1):
        print("*", end=" ")
    print()
```

Output :

```
        *
      * * *
    * * * * *
  * * * * * * *
* * * * * * * * *
```

Pattern 4 : inverted

```
rows = 5
for i in range(rows, 0, -1):
    for j in range(rows - i):
        print(" ", end=" ")
    for k in range(2 * i - 1):
        print("*", end=" ")
    print()
```

Output :

```
* * * * * * * * *
  * * * * * * *
    * * * * *
      * * *
        *
```

Star Patterns: Part 2 & Practice

Decide the rows first, then count the spaces and stars in each row.

1) Right-aligned triangle

Example :

```
rows = 5
for i in range(1, rows + 1):
    for j in range(rows - i):
        print(" ", end=" ")
    for k in range(i):
        print("*", end=" ")
    print()
```

Output :

```
        *
      * *
    * * *
  * * * *
* * * * *
```

Which loops?

Pattern	Space loop	Star loop
Increasing triangle	none	range(i)
Decreasing triangle	none	range(i), rows count down
Centered pyramid	range(rows - i)	range(2 * i - 1)
Inverted pyramid	range(rows - i), rows count down	range(2 * i - 1)
Right-aligned triangle	range(rows - i)	range(i)

2) Practice

Task 1 Racecar palindrome?

Solution :

```
word = "racecar"
if word == word[::-1]:
    print("Palindrome")
else:
    print("Not a palindrome")
```

```
Palindrome
```

Task 2 Number palindrome (input).

Solution :

```
number = input("Enter a number: ")
if number == number[::-1]:
    print("Palindrome")
else:
    print("Not a palindrome")
```

```
Enter a number: 12321
Palindrome
```

Task 3 Pyramid with 4 rows.

Solution :

```
rows = 4
for i in range(1, rows + 1):
    for j in range(rows - i):
        print(" ", end=" ")
    for k in range(2 * i - 1):
        print("*", end=" ")
    print()
```

```
    *
  * * *
* * * * *
* * * * * * *
```

Task 4 Right-aligned triangle, 4 rows.

Solution :

```
rows = 4
for i in range(1, rows + 1):
    for j in range(rows - i):
        print(" ", end=" ")
    for k in range(i):
        print("*", end=" ")
    print()
```

```
    *
  * *
* * *
* * * *
```